CSC3209 - Software Architecture and Design Pattern – Tutorial 5

1- Assume that you work as a software quality manager. As part of your job, you are required to specify different scenarios of different quality attributes. This is to manage and enhance quality of the system handled by you. Present different scenarios for the following quality attributes

- Availability
- Modifiability
- Performance
- Security
- Testability
- Usability

Employ your knowledge you learned in the lecture to present the quality attributes scenarios.

Solution:

Availability: Availability refers to a property of software that it is there and ready to carry out its task when you need it to be. Availability as an aspect of dependability is defined as: “Availability refers to the ability of a system to mask or repair faults such that the cumulative service outage period does not exceed a required value over a specified time interval.

Table 1 provides examples of system availability requirements and associated threshold values for acceptable system downtime. Table 2 presents availability general scenario.

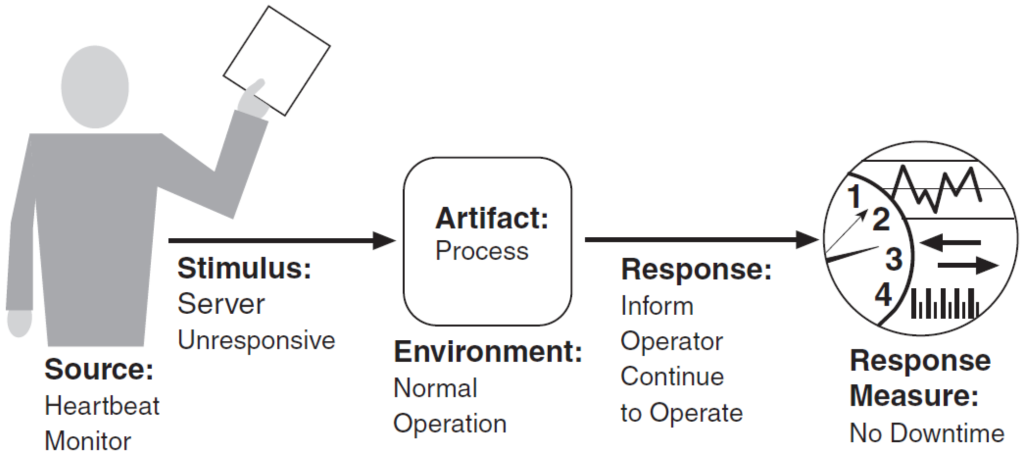
-Table1-

Availability	Downtime/90 Days	Downtime/Year
99.0%	21 hours, 36 minutes	3 days, 15.6 hours
99.9%	2 hours, 10 minutes	8 hours, 0 minutes, 46 seconds
99.99%	12 minutes, 58 seconds	52 minutes, 34 seconds
99.999%	1 minute, 18 seconds	5 minutes, 15 seconds
99.9999%	8 seconds	32 seconds

-Table2-

Portion of Scenario	Possible Values
Source	Internal/external: people, hardware, software, physical infrastructure, physical environment
Stimulus	Fault: omission, crash, incorrect timing, incorrect response
Artifact	Processors, communication channels, persistent storage, processes
Environment	Normal operation, startup, shutdown, repair mode, degraded operation, overloaded operation
Response	Prevent the fault from becoming a failure Detect the fault: ▪ Log the fault ▪ Notify appropriate entities (people or systems) Recover from the fault: ▪ Disable source of events causing the fault ▪ Be temporarily unavailable while repair is being effected ▪ Fix or mask the fault/failure or contain the damage it causes ▪ Operate in a degraded mode while repair is being effected
Response Measure	Time or time interval when the system must be available Availability percentage (e.g., 99.999%) Time to detect the fault Time to repair the fault Time or time interval in which system can be in degraded mode Proportion (e.g., 99%) or rate (e.g., up to 100 per second) of a certain class of faults that the system prevents, or handles without failing

Figure 1 shows a concrete scenario generated from the general scenario: The heartbeat monitor determines that the server is nonresponsive during normal operations. The system informs the operator and continues to operate with no downtime.



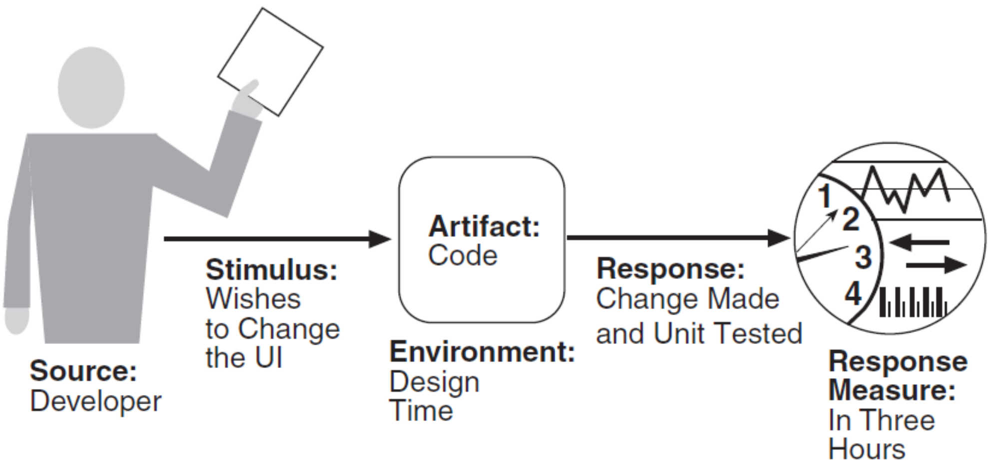
-Figure1-

Modifiability: Modifiability is about change, and our interest in it centers on the cost and risk of making changes. Table3 presents modifiability general scenario.

-Table3-

Portion of Scenario	Possible Values
Source	End user, developer, system administrator
Stimulus	A directive to add/delete/modify functionality, or change a quality attribute, capacity, or technology
Artifacts	Code, data, interfaces, components, resources, configurations, . . .
Environment	Runtime, compile time, build time, initiation time, design time
Response	One or more of the following: ▪ Make modification ▪ Test modification ▪ Deploy modification
Response Measure	Cost in terms of the following: ▪ Number, size, complexity of affected artifacts ▪ Effort ▪ Calendar time ▪ Money (direct outlay or opportunity cost) ▪ Extent to which this modification affects other functions or quality attributes ▪ New defects introduced

Figure 2 illustrates a concrete modifiability scenario: The developer wishes to change the user interface by modifying the code at design time. The modifications are made with no side effects within three hours.



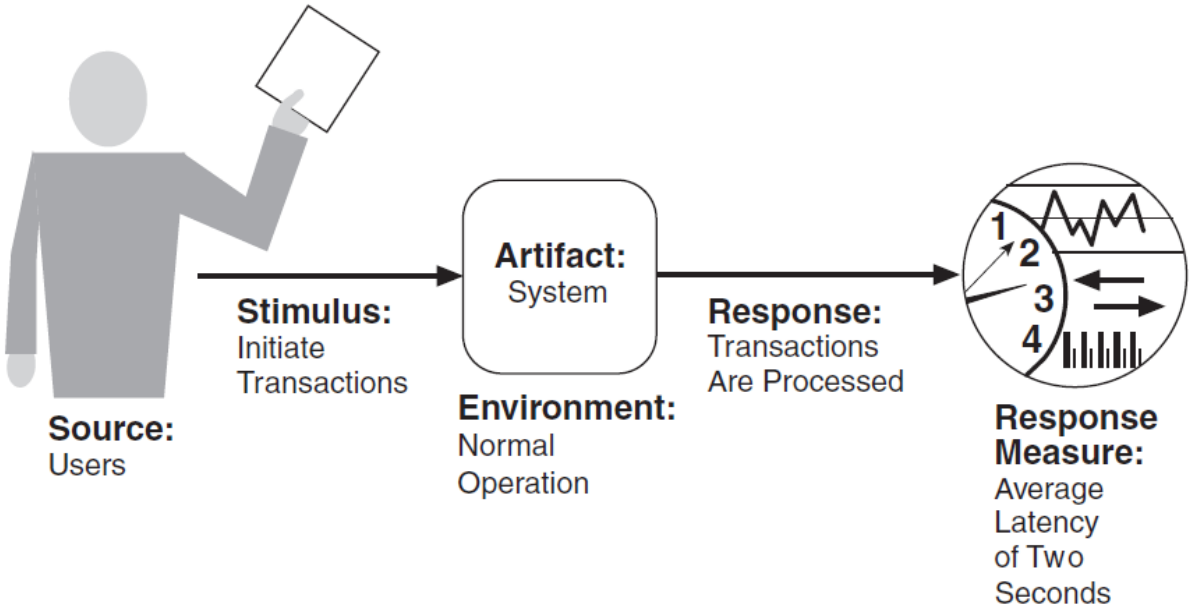
-Figure2-

Performance: Performance is about timing. Events (interrupts, messages, requests from users, or the passage of time) occur, and the system must respond to them. One of the things that make performance complicated is the number of event sources and arrival patterns. Table 4 presents the performance general scenario.

Figure 3 gives an example concrete performance scenario: Users initiate transactions under normal operations. The system processes the transactions with an average latency of two seconds.

-Table 4-

Portion of Scenario	Possible Values
Source	Internal or external to the system
Stimulus	Arrival of a periodic, sporadic, or stochastic event
Artifact	System or one or more components in the system
Environment	Operational mode: normal, emergency, peak load, overload
Response	Process events, change level of service
Response Measure	Latency, deadline, throughput, jitter, miss rate



-Figure3-

Security: Security is a measure of the system's ability to protect data and information from unauthorized access while still providing access to people and systems that are authorized.

Security can be characterized as a system providing nonrepudiation, confidentiality, integrity, assurance, availability, and auditing.

The simplest approach to characterizing security has three characteristics: confidentiality, availability, integrity and availability.

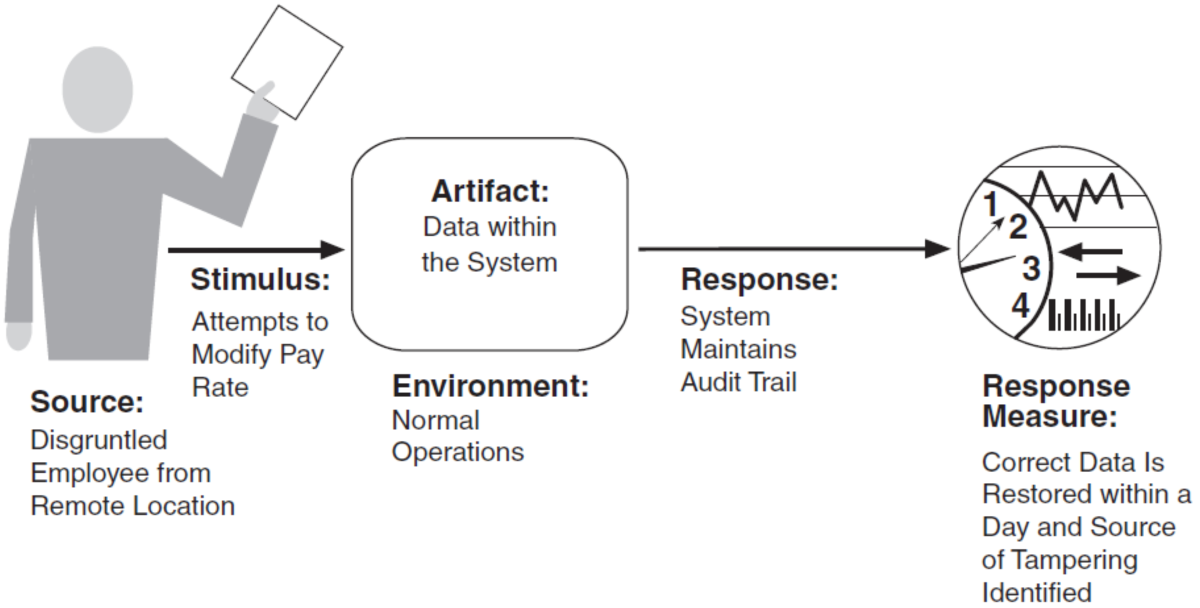
- Confidentiality is the property that data or services are protected from unauthorized access. For example, a hacker cannot access your income tax returns on a government computer.
- Integrity is the property that data or services are not subject to unauthorized manipulation. For example, your grade has not been changed since your instructor assigned it.
- Availability is the property that the system will be available for legitimate use. For example, a denial-of-service attack won't prevent you from ordering a book from an online bookstore.
- Authentication verifies the identities of the parties to a transaction and checks if they are truly who they claim to be. For example, when you get an e-mail purporting to come from a bank, authentication guarantees that it actually comes from the bank.

- Nonrepudiation guarantees that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message. For example, you cannot deny ordering something from the Internet, or the merchant cannot disclaim getting your order.
- Authorization grants a user the privileges to perform a task. For example, an online banking system authorizes a legitimate user to access his account.

Table5 presents the general scenario of security. Figure 9.1 shows a sample concrete scenario: A disgruntled employee from a remote location attempts to modify the pay rate table during normal operations. The system maintains an audit trail, and the correct data is restored within a day.

-Table 5-

Portion of Scenario	Possible Values
Source	Human or another system which may have been previously identified (either correctly or incorrectly) or may be currently unknown. A human attacker may be from outside the organization or from inside the organization.
Stimulus	Unauthorized attempt is made to display data, change or delete data, access system services, change the system's behavior, or reduce availability.
Artifact	System services, data within the system, a component or resources of the system, data produced or consumed by the system
Environment	The system is either online or offline; either connected to or disconnected from a network; either behind a firewall or open to a network; fully operational, partially operational, or not operational.
Response	Transactions are carried out in a fashion such that ▪ Data or services are protected from unauthorized access. ▪ Data or services are not being manipulated without authorization. ▪ Parties to a transaction are identified with assurance. ▪ The parties to the transaction cannot repudiate their involvements. ▪ The data, resources, and system services will be available for legitimate use. The system tracks activities within it by ▪ Recording access or modification ▪ Recording attempts to access data, resources, or services ▪ Notifying appropriate entities (people or systems) when an apparent attack is occurring
Response Measure	One or more of the following: ▪ How much of a system is compromised when a particular component or data value is compromised ▪ How much time passed before an attack was detected ▪ How many attacks were resisted ▪ How long does it take to recover from a successful attack ▪ How much data is vulnerable to a particular attack



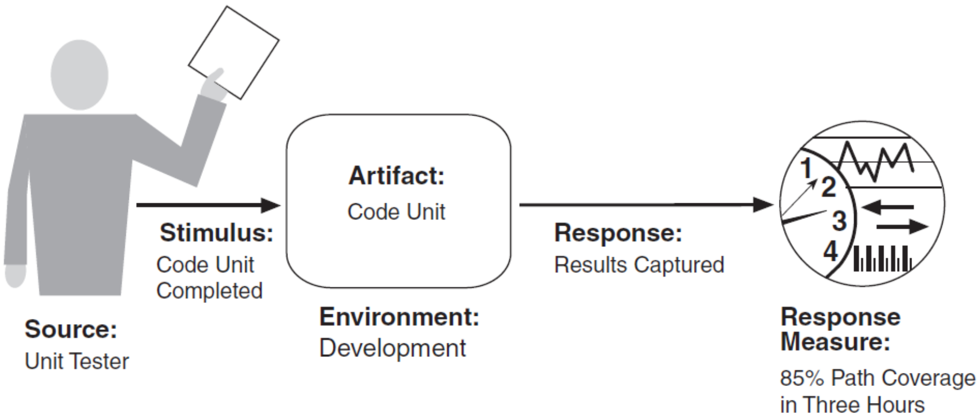
-Figure4-

Testability: Software testability refers to the ease with which software can be made to demonstrate its faults through (typically execution-based) testing. If a fault is present in a system, then we want it to fail during testing as quickly as possible. For a system to be properly testable, it must be possible to control each component's inputs (and possibly manipulate its internal state) and then to observe its outputs (and possibly its internal state).

Table6 presents the general scenario of testability. Figure 5 shows a concrete scenario for testability. The unit tester completes a code unit during development and performs a test sequence whose results are captured and that gives 85 percent path coverage within three hours of testing.

-Table 6-

Portion of Scenario	Possible Values
Source	Unit testers, integration testers, system testers, acceptance testers, end users, either running tests manually or using automated testing tools
Stimulus	A set of tests is executed due to the completion of a coding increment such as a class layer or service, the completed integration of a subsystem, the complete implementation of the whole system, or the delivery of the system to the customer.
Environment	Design time, development time, compile time, integration time, deployment time, run time
Artifacts	The portion of the system being tested
Response	One or more of the following: execute test suite and capture results, capture activity that resulted in the fault, control and monitor the state of the system
Response Measure	One or more of the following: effort to find a fault or class of faults, effort to achieve a given percentage of state space coverage, probability of fault being revealed by the next test, time to perform tests, effort to detect faults, length of longest dependency chain in test, length of time to prepare test environment, reduction in risk exposure (size(loss) × prob(loss))



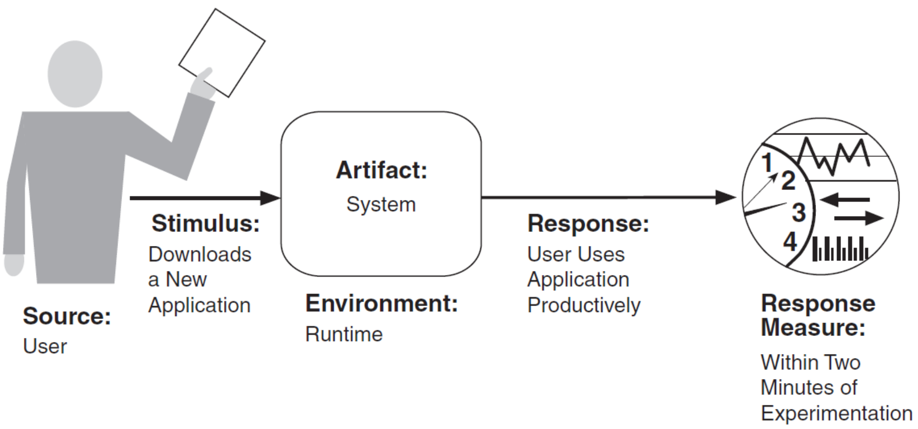
-Figure5-

Usability: Usability is concerned with how easy it is for the user to accomplish a desired task and the kind of user support the system provides. Table 7 enumerates the elements of the general scenario that characterize usability.

-Table 7-

Portion of Scenario	Possible Values
Source	End user, possibly in a specialized role
Stimulus	End user tries to use a system efficiently, learn to use the system, minimize the impact of errors, adapt the system, or configure the system.
Environment	Runtime or configuration time
Artifacts	System or the specific portion of the system with which the user is interacting
Response	The system should either provide the user with the features needed or anticipate the user's needs.
Response Measure	One or more of the following: task time, number of errors, number of tasks accomplished, user satisfaction, gain of user knowledge, ratio of successful operations to total operations, or amount of time or data lost when an error occurs

Figure 6 gives an example of a concrete usability scenario that you could generate using Table 7: The user downloads a new application and is using it productively after two minutes of experimentation.



-Figure6-